

SWARALI DESHMUKH

Data Scientist | AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with **1+** years of applied experience across **econometric modelling**, **NLP systems**, **computer vision**, and **full-stack ML deployment**. Proven track record at **WRI** and **ISRO** delivering models that informed **climate and energy policy decisions** on **1TB+ datasets**. Built production-grade systems including a **LangGraph agentic pipeline** with **Gemini 2.5**, a **contrastive learning** fine-tuned **sentence transformer** (Pearson r +36%), and **Dockerized full-stack NLP applications**. Comfortable owning projects end-to-end — from **raw data pipelines** to **deployed REST APIs** and **React frontends**.

TECHNICAL SKILLS

Languages: Python, R

ML / Classical: Scikit-learn, XGBoost, Random Forest, Kernel Ridge Regression, Word2Vec, Hyperparameter Tuning (GridSearchCV)

Deep Learning: PyTorch, TensorFlow, CNNs, Transformers, BERT, Sentence Transformers (all-MiniLM-L6-v2), Retrieval Augmented Generation

NLP & GenAI: RAG Pipelines, LangChain, LangGraph, BERT Embeddings, Named Entity Recognition, Text Classification, Contrastive Learning, LLM APIs (Gemini)

Data Engineering: Pandas, NumPy, ETL Pipelines, Chunked Processing, Geospatial Data (GDAL), Econometric Modelling

Infrastructure & Web: FastAPI, React, Docker, Docker Compose, REST APIs, PostgreSQL, MongoDB, Git

WORK EXPERIENCE

CEF-Intern | World Resources Institute (WRI)

Climate, Economics & Finance

Bangalore

Jun 2026 – present

- **Ascent Tool:** Debugged and refactored a VBA-based automation system, resolving core stability issues and consolidating UI logic into a unified control layer for production use.
- **IDEEA Model (Environment Setup):** Configured and validated a multi-package R modeling framework from source, troubleshooting dependency and build issues to enable reliable local deployment.
- **IDEEA Model (Data Collection):** Built a structured dataset from fragmented industry sources to feed energy-systems modeling and scenario analysis.

Junior Program Associate | World Resources Institute (WRI)

Climate, Economics & Finance

Remote

Apr 2025 – Jul 2025

- Built **econometric impact models** in **Python** and **R** to evaluate **distributional effects of carbon pricing policies** across income quintiles using **NSSO** and **HCES** household survey data, adopted by WRI's India Climate Finance team.
- Designed and deployed **automated ETL pipelines** that cut **data cleaning time by ~60%**, enabling analysts to run policy scenario comparisons **3× faster**.

Python Developer Intern | World Resources Institute (WRI)
Climate, Economics & Finance

Remote
Jul 2024 – Jan 2025

- Migrated the **FTT energy model** from **Fortran to Python**, reducing codebase complexity by **~40%** and enabling **parallel simulation runs**.
- Built **memory-efficient data pipelines** handling **HCES datasets** using **chunked processing**.
- Refactored **5+ legacy modeling modules** into well-documented, reusable **Python components** with **unit tests**.

Machine Learning Intern | Space Applications Centre (ISRO)
Remote Sensing & Geospatial AI

Ahmedabad
Jan 2024 – Apr 2024

- Processed and fused **multispectral satellite imagery** with tabular agricultural datasets to train **crop yield prediction models** ($R^2 = 0.84$).
- Implemented **Kernel Ridge Regression** with **GridSearchCV**, improving forecast accuracy by **~18%** over baseline.
- Built **scalable preprocessing pipelines** integrating **geospatial** and tabular data (reduced time from **4h to <45min**).

AI Intern | Anantadi Tech Pvt Ltd
Computer Vision & Ad-Tech

Pune
Jun 2023 – Aug 2023

- Built and evaluated object detection pipelines using **YOLOv8** and **Detectron2** for identifying context-aware ad placement regions (walls, flat surfaces, banners) in movie and TV video frames.
- Benchmarked object tracking algorithms (**DLIB**, **Norfair**, **GOTURN**) for maintaining spatial consistency of ad placement regions across video frames.
- Operated in an **Agile** sprint framework with twice-daily scrums, presenting weekly progress updates and technical findings to company leadership and faculty advisors.

PROJECTS

CareerWise-AI — AI Recruitment Recommendation System

- Built a full-stack AI-powered resume ranking system using **FastAPI**, **React (Vite)**, and a fine-tuned **Sentence-BERT (MiniLM-L6-v2)** model to compute **semantic similarity** scores between resumes and job descriptions, containerized with **Docker** and **nginx** reverse proxy.
- Developed an **LLM-driven** career path advisor leveraging **LangGraph**, **Google Gemini API**, and **LangChain** to parse resumes (**PDF/DOCX**), detect **skill gaps**, and generate personalized course, job role, and university recommendations with real-time job listings via the **Adzuna API**.
- Designed and deployed a **REST API** with **FastAPI/Uvicorn** featuring explainability outputs (**sentence-level scoring**, **occlusion analysis**, **keyword gap detection**), **MongoDB** persistence, automated file lifecycle management, and **CORS**-secured multi-origin support for production environments.

Emotion Detection from Audio

- Built a **speech emotion classifier** using **MFCC**, **pitch**, and **spectral features** from **RAVDESS** + **TESS** datasets (**81% accuracy**).
- Designed a **modular inference pipeline** with **<200ms latency**.

EDUCATION

M.Sc. Artificial Intelligence & Machine Learning
Vellore Institute of Technology, Vellore

Jul 2025 – Expected Aug 2027
SGPA: 9.46 / 10

B.Sc. Data Science

Christ (Deemed to be University), Pune

2021 – 2024

CGPA: 9.2 / 10

- **Extra-Curricular Excellence Merit Award** (2 consecutive years)
- **Top 10 ranking** — Data Analysis Project: Happiness Index via Twitter Data